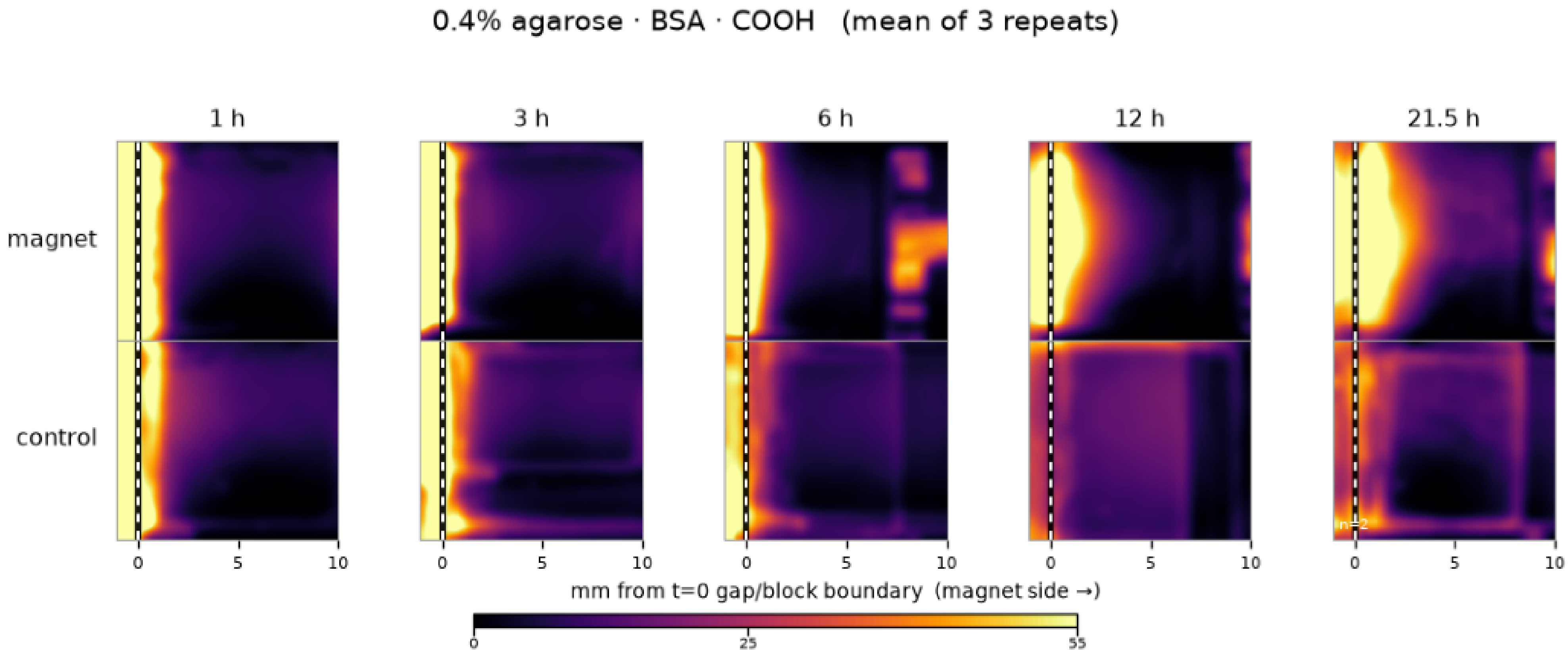
Magnetic Nanoparticle Transport in Agarose

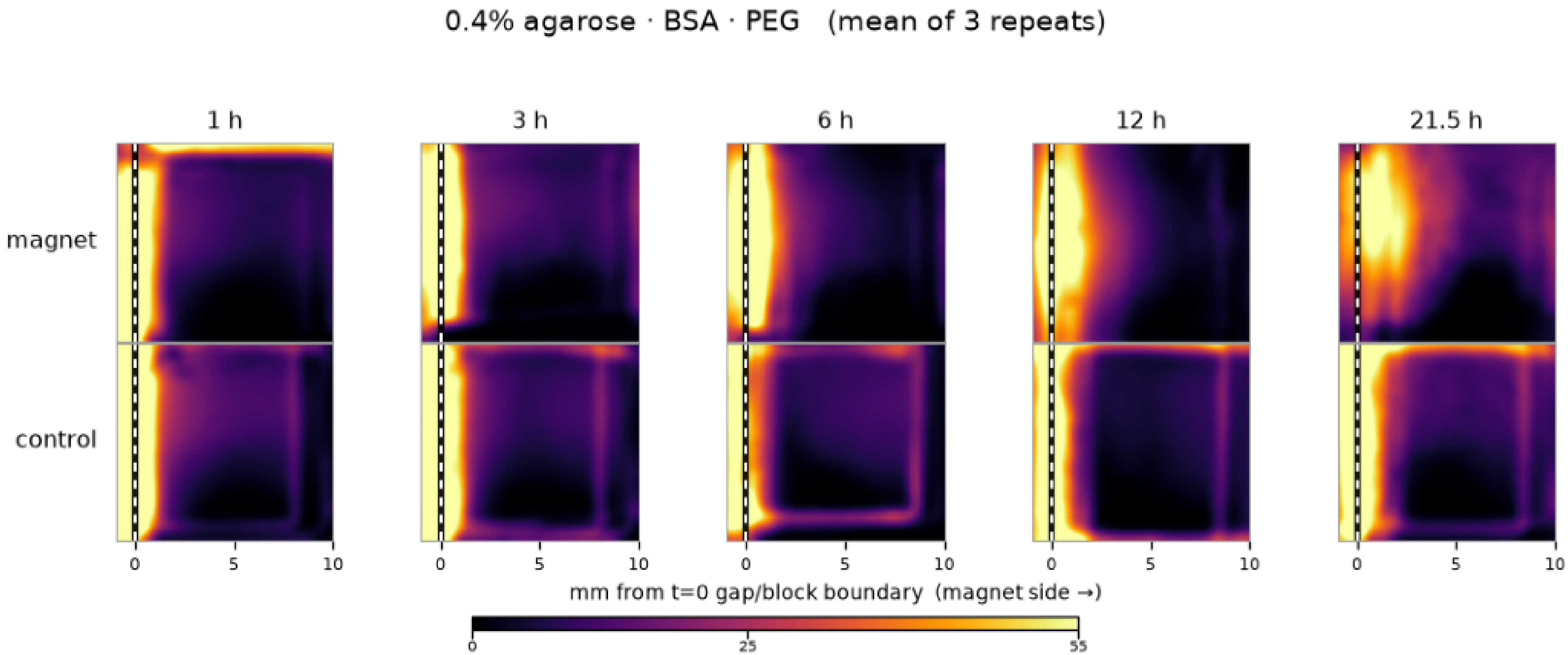
Block heatmaps — 21.5 h back-injection run, 27 Aug 2026

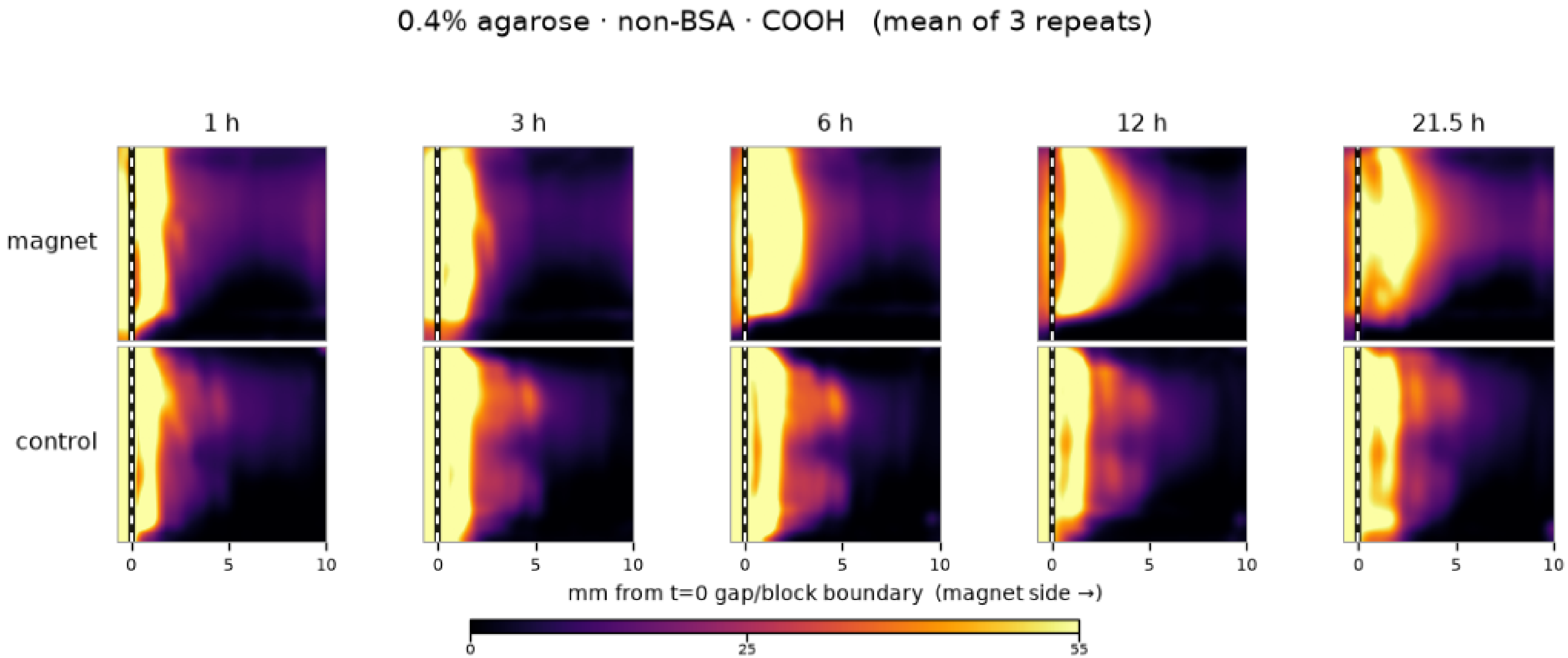
Each page shows one condition: the magnet arm (top row) against its no-magnet controls (bottom row), each cell the mean of 3 repeats, at 1, 3, 6, 12 and 21.5 hours. Every cell is the 10×10 mm agarose block seen from above — injection gap at the left, magnet side at the right. Color is NP darkening above each frame's own gel floor (camera exposure cancelled); the white dashed line marks the gap/block boundary measured at t = 0 and held fixed (agarose shrinkage is shown, not compensated). One shared color scale (0–55 L*) across all pages, so any two cells are directly comparable.

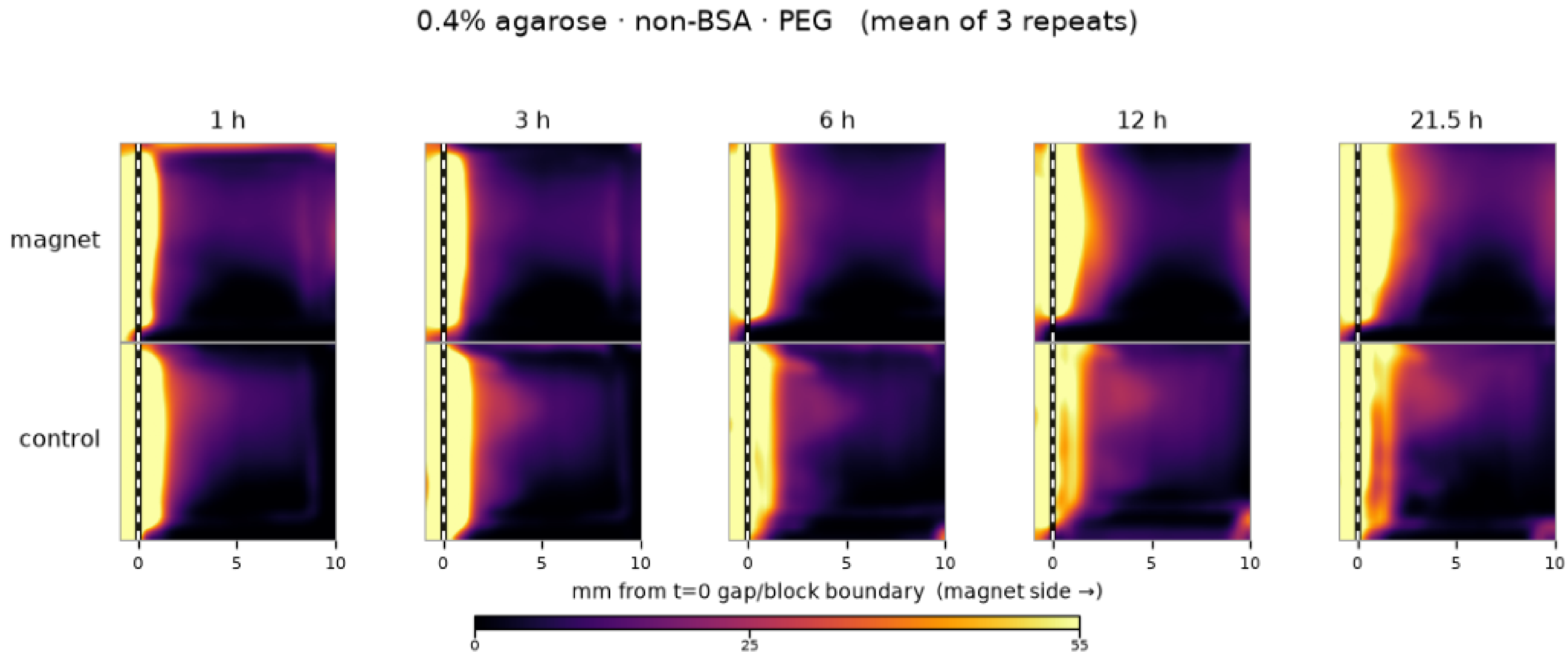
Frames are registered onto their series' t = 0 image and anchored on the case's rim edge and the injection gap. In magnet rows, the magnet's shadow adds some signal toward the right edge (deepening with ambient light changes; the 12 h photos were taken at night) — compare against the control row beneath.

github.com/sthoa/nanoparticles

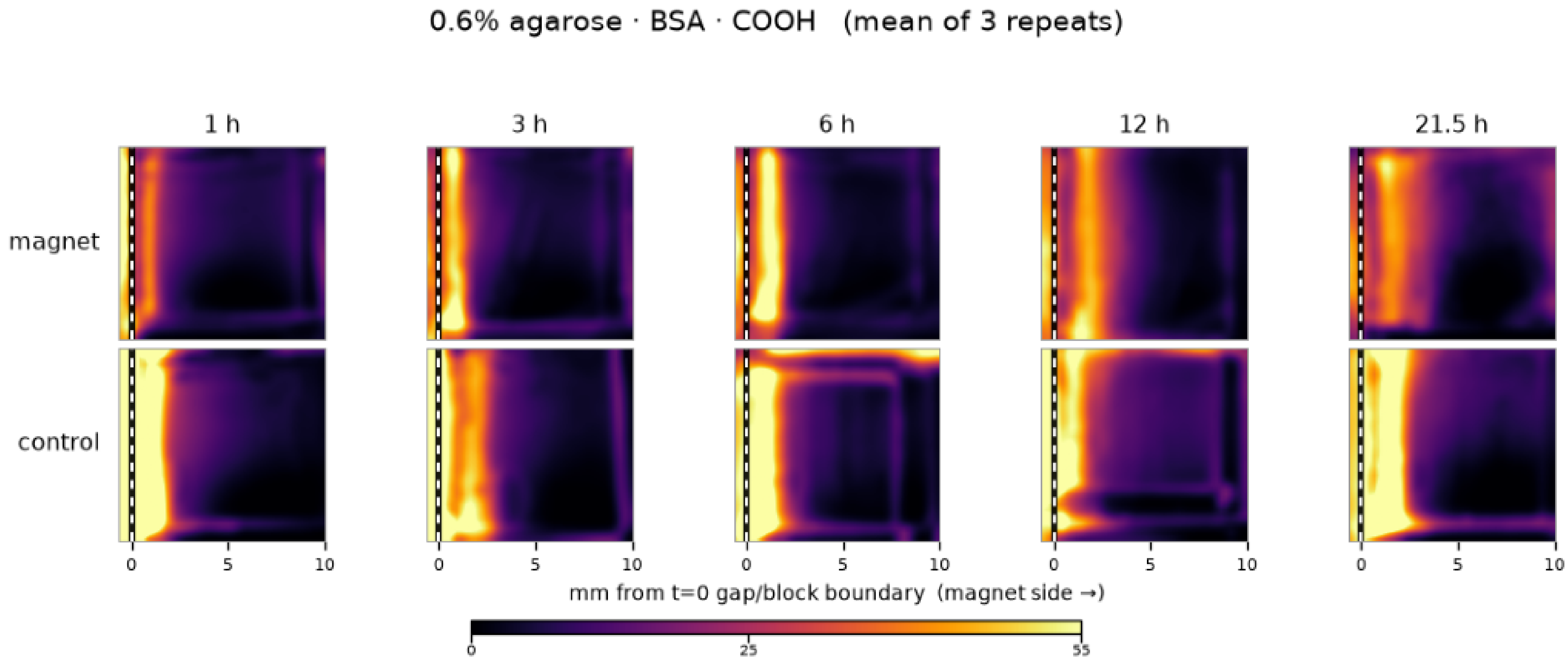

0.4% agarose · BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
n=2
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.4% agarose · BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55

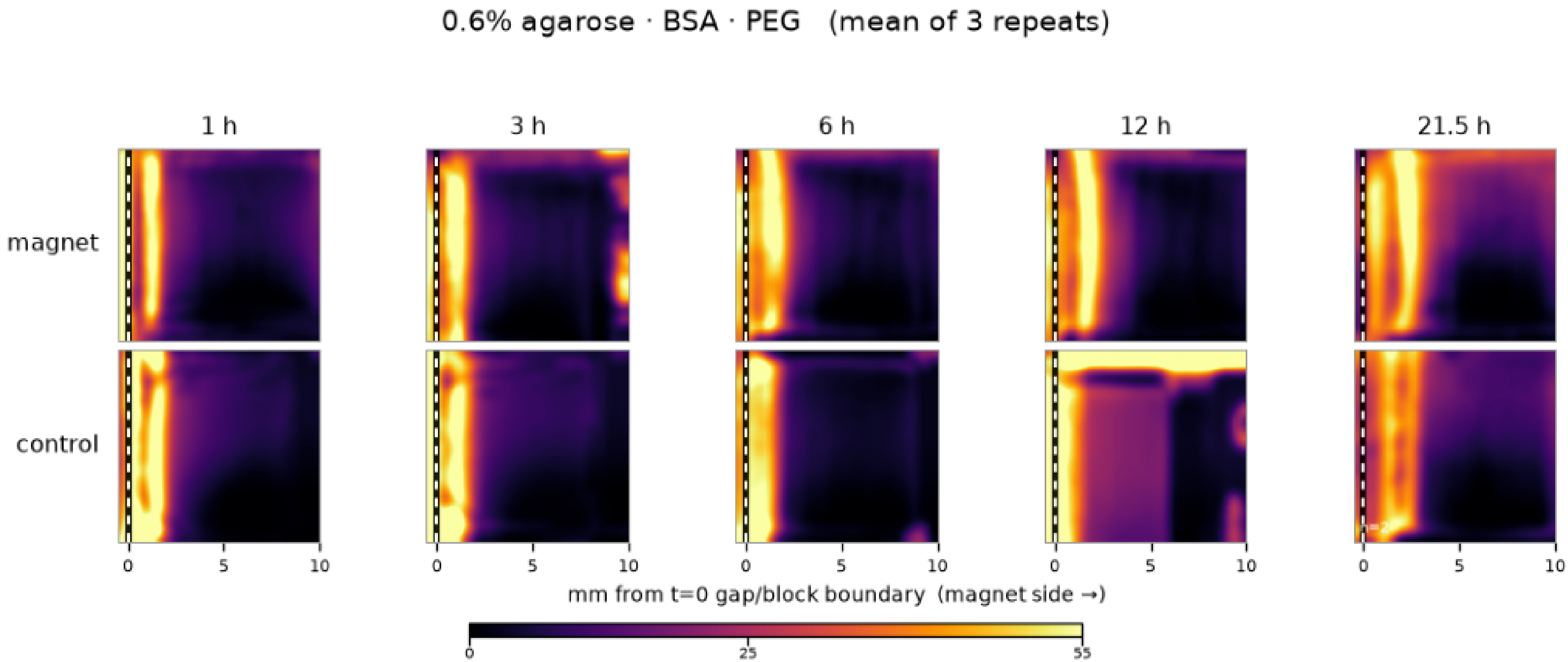

0.4% agarose · non-BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55

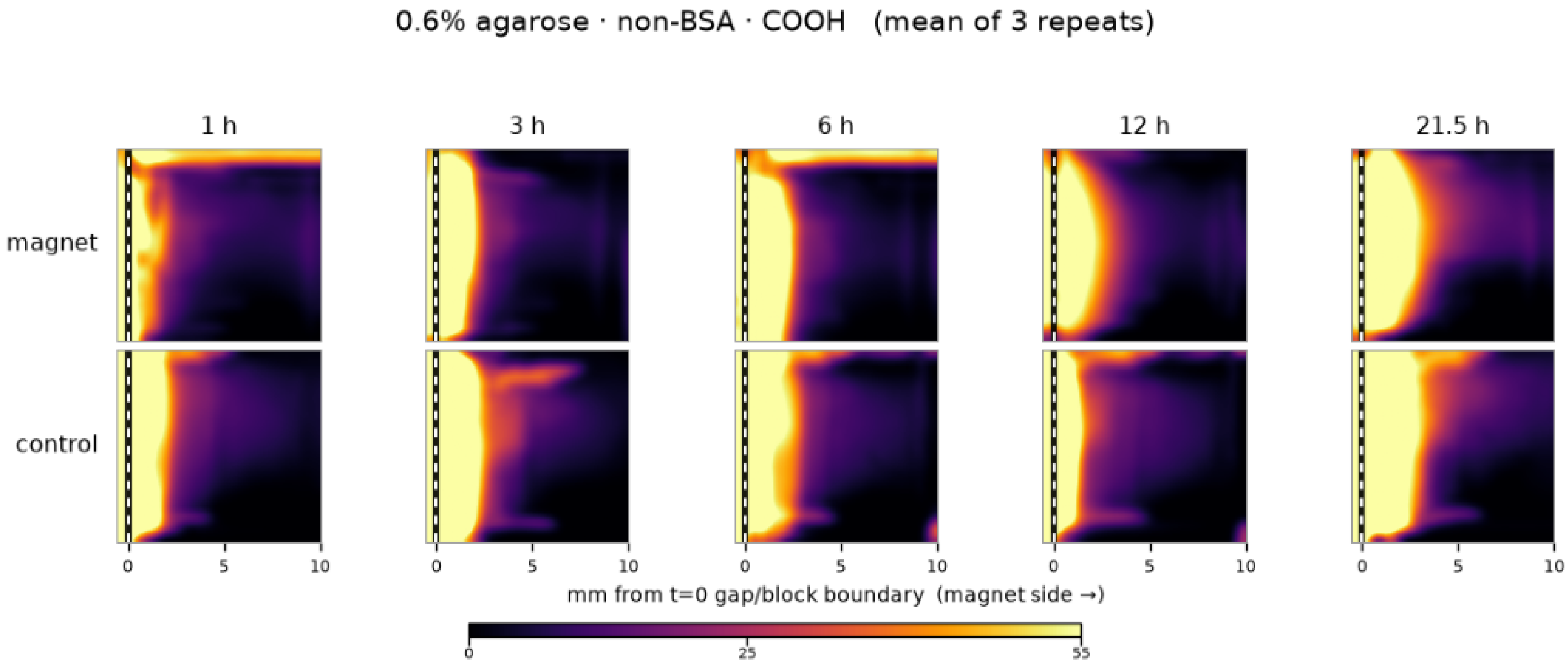


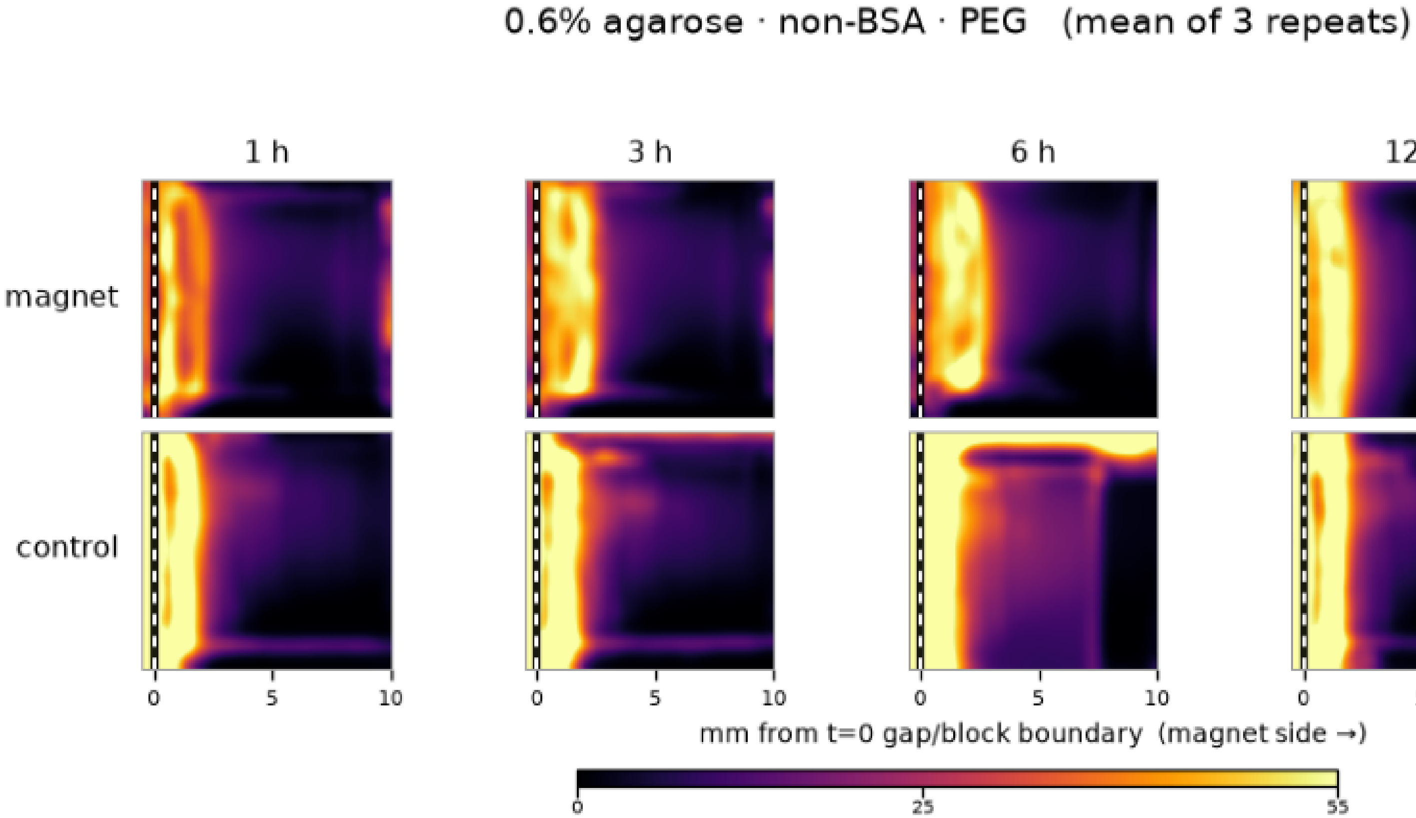
0.4% agarose · non-BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55



0.6% agarose · BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.6% agarose · BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.6% agarose · non-BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.6% agarose · non-BSA · PEG   (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55